IoT & Embedded Systems Portfolio

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About Me

Electrical and Electronics Engineering graduate with hands-on experience in embedded systems, IoT applications, and control systems using Arduino and ESP8266. Passionate about applying smart technologies to energy systems, automation, and real-world engineering challenges. Long-term goal is to contribute to renewable energy development and smart grid modernization.

Technical Skills

- Arduino Programming (C/C++)
- Embedded Systems Design
- IoT Systems (ESP8266, Blynk, IFTTT)
- Circuit Design & Prototyping
- Sensors & Actuators Integration
- Control Logic Systems
- Basic Networking (WiFi Communication)

PROJECTS

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1. Digital Piano System

Objective

Design a simple digital piano using Arduino to generate tones based on user input.

Key Learning

- Signal generation
- Human-device interaction

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2. Digital Thermometer

Objective

Measure and display temperature using sensor integration.

Key Learning

- Analog-to-digital conversion
- Sensor calibration

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3. Encrypted Safe System

Objective

Develop a password-protected locking system using Arduino.

Key Learning

- Security logic implementation
- Keypad interfacing

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4. Automatic Door System

Objective

Create an automated door using sensors and actuators.

Key Learning

- Motion detection
- Actuator control

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5. Digital Clock

Objective

Design a real-time digital clock system.

Key Learning

- Time-based programming

- Display interfacing

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6. Parking Sensor System

Objective

Detect distance and provide feedback for parking assistance.

Key Learning

- Ultrasonic sensing
- Distance measurement systems

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7. Bluetooth Controlled Car

Objective

Control a robotic vehicle wirelessly via Bluetooth.

Key Learning

- Wireless communication
- Motor control systems

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8. Obstacle Avoiding Robot

Objective

Develop a robot capable of avoiding obstacles autonomously.

Key Learning

- Autonomous navigation
- Real-time decision making

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9. Line Following Robot

Objective

Design and implement an autonomous robot capable of following a predefined path using sensor feedback.

Key Learning

- Feedback control systems
- Sensor-based automation

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10. Smart IoT Socket

Objective

Develop a smart socket that allows remote control of electrical devices via the internet.

Key Learning

- IoT system architecture
- Safe AC switching using relays

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11. Smart Lamp with Voice Control

Objective

Control electrical devices using mobile app and voice commands via cloud platforms.

Key Learning

- Cloud integration
- Voice-controlled automation

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12. AI-Based Voice Controlled RGB LED

Objective

Control LED colors using voice recognition and AI blocks.

Key Learning

- Human-machine interaction
- AI integration in embedded systems

Project Summary & Progression

This project series demonstrates a structured progression from basic embedded systems (input/output control, sensors, and displays) to advanced applications involving robotics, wireless communication, Internet of Things (IoT), and artificial intelligence integration.

Future Work

- Integration with solar energy systems
- Smart energy monitoring solutions
- Development of scalable smart grid applications

Conclusion

This portfolio demonstrates foundational to intermediate experience in embedded systems and IoT, with clear progression toward intelligent automation and energy systems applications.

Contact

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